

Program 10: Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

Step 1: Start Minikube with required resources

```
minikube start --cpus=2 --memory=4096
```

Step 2: Enable and open Kubernetes Dashboard

```
minikube dashboard
```

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ minikube dashboard
🔍 Verifying dashboard health ...
🚀 Launching proxy ...
🔍 Verifying proxy health ...
🌐 Opening http://127.0.0.1:38129/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Gtk-Message: 15:43:36.424: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^C
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🔍 Using the docker driver based on existing profile
⚠️ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
📦 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
  ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
  ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard, metrics-server
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 3: Create project directory

```
mkdir Program-10
```

```
cd Program-10
```

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ mkdir program10
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ cd program10
```

Step 4: Create Deployment YAML file

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ nano deployment.yaml
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ nano service.yaml
```

```
GNU nano 7.2 deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 5: Service YAML file

Create

```
GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 6: Apply Deployment and Service

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 7: Check running Pods

kubectl get pods

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
hello-cron-29435655-n7rg4	0/1	Completed	0	2m56s
hello-cron-29435656-hpmjc	0/1	Completed	0	2m16s
hello-cron-29435657-4tl6n	0/1	Completed	0	76s
hello-cron-29435658-57j84	0/1	ContainerCreating	0	16s
nginx-deployment-6f9664446b-2mw2x	1/1	Running	0	18s
nginx-deployment-6f9664446b-2tp8d	1/1	Running	0	18s

Step 8: Check Kubernetes Nodes

kubectl get nodes

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane	23d	v1.34.0

Step 9: Check Services

kubectl get svc

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ kubectl get service
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	6d23h
nginx-service	NodePort	10.109.192.150	<none>	80:32692/TCP	29s

Step 10: View Pod Logs

kubectl logs nginx-deployment-6f9664446b-86rj6

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ kubectl logs nginx-deployment-6f9664446b-2mw2x
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 10:18:14 [notice] 1#1: using the "epoll" event method
2025/12/19 10:18:14 [notice] 1#1: nginx/1.29.4
2025/12/19 10:18:14 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/19 10:18:14 [notice] 1#1: OS: Linux 6.14.0-35-generic
2025/12/19 10:18:14 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/19 10:18:14 [notice] 1#1: start worker processes
2025/12/19 10:18:14 [notice] 1#1: start worker process 29
2025/12/19 10:18:14 [notice] 1#1: start worker process 30
2025/12/19 10:18:14 [notice] 1#1: start worker process 31
2025/12/19 10:18:14 [notice] 1#1: start worker process 32
2025/12/19 10:18:14 [notice] 1#1: start worker process 33
2025/12/19 10:18:14 [notice] 1#1: start worker process 34
2025/12/19 10:18:14 [notice] 1#1: start worker process 35
2025/12/19 10:18:14 [notice] 1#1: start worker process 36
2025/12/19 10:18:14 [notice] 1#1: start worker process 37
2025/12/19 10:18:14 [notice] 1#1: start worker process 38
2025/12/19 10:18:14 [notice] 1#1: start worker process 39
2025/12/19 10:18:14 [notice] 1#1: start worker process 40
```

Step 11: Describe Pod Details

kubectl describe pod nginx-deployment-6f9664446b-86rj6

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ kubectl describe pod nginx-deployment-6f9664446b-2mw2x
Name:          nginx-deployment-6f9664446b-2mw2x
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Fri, 19 Dec 2025 15:47:58 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.1.49
IPs:
  IP:          10.244.1.49
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://5f92f38115b33e9a26c009def42bc4ac85693560779e2eeadf326c011abf296a
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:         80/TCP
    Host Port:    0/TCP
    State:        Running
      Started:    Fri, 19 Dec 2025 15:48:14 +0530
    Ready:        True
    Restart Count: 0
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-lb9xs (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                       True
  Ready                             True
  ContainersReady                   True
  PodScheduled                      True
Volumes:
  kube-api-access-lb9xs:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
```

Step 12: Access Application via Minikube

minikube service nginx-service

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32692

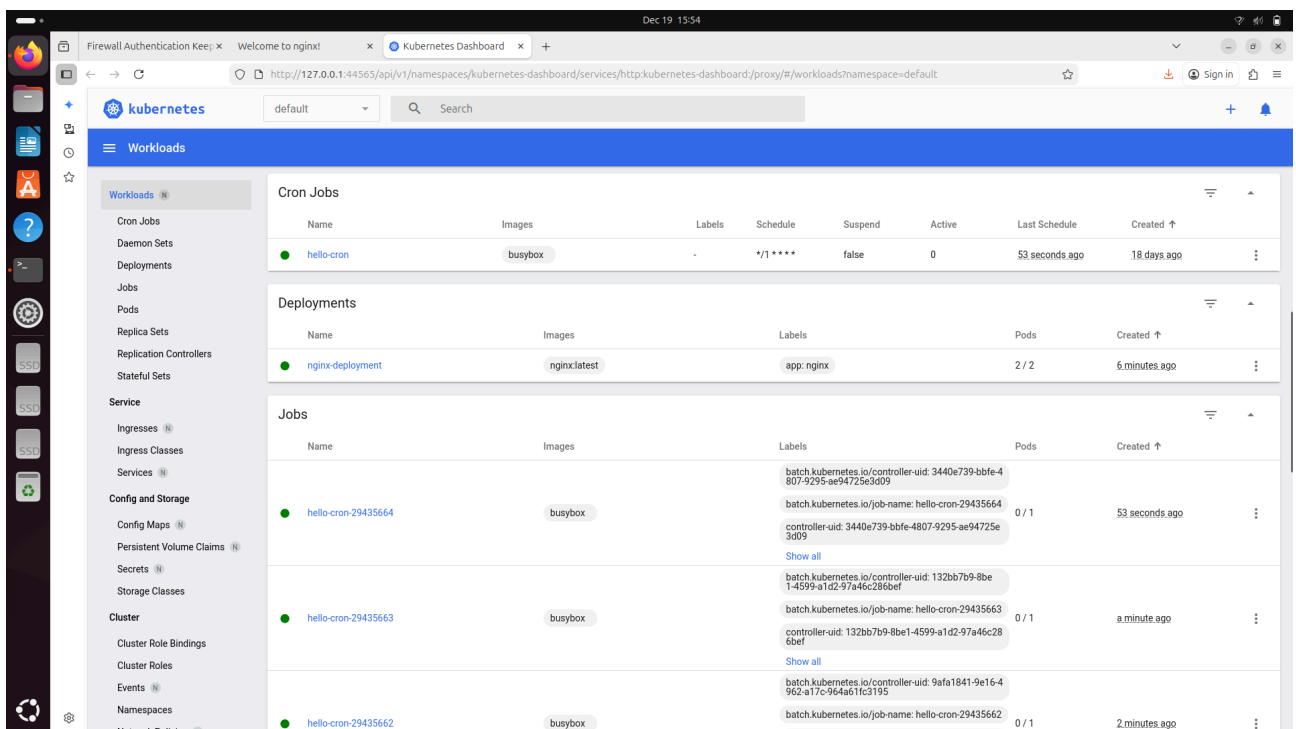
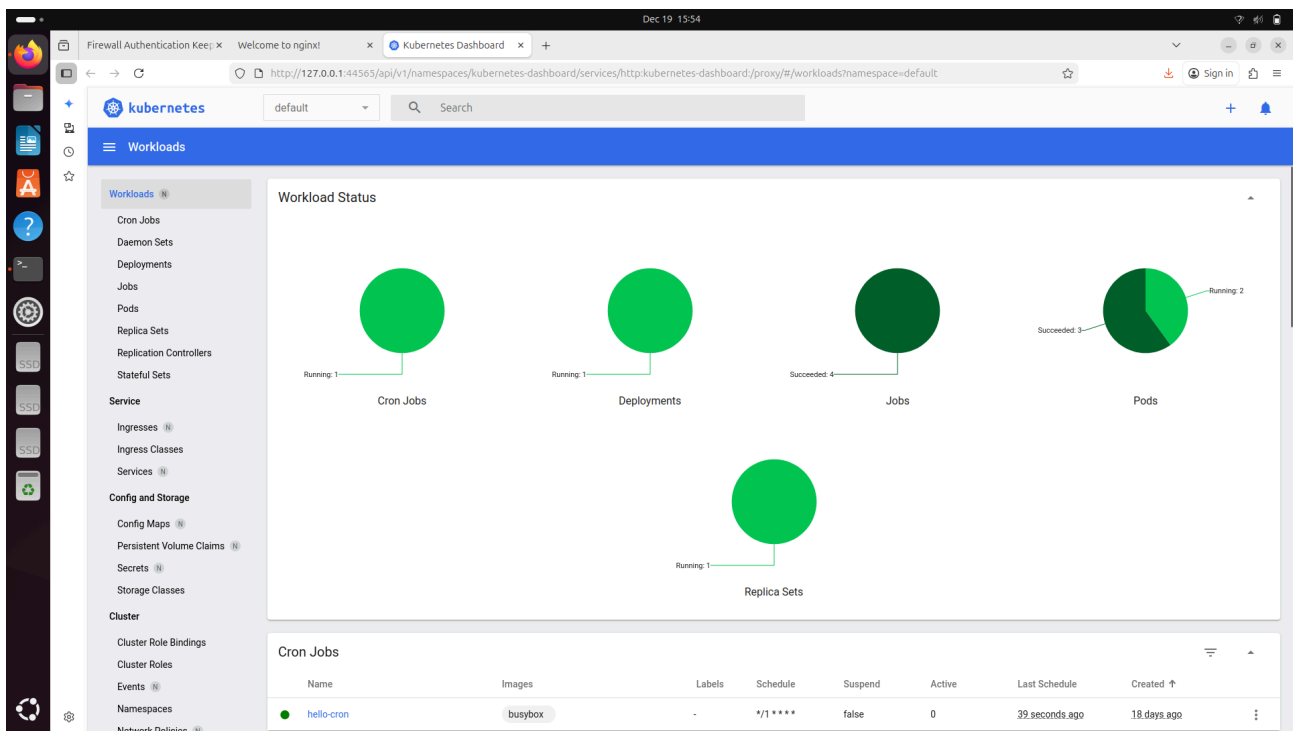
```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ minikube service nginx-service
Opening service default/nginx-service in default browser...
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program10$ minikube service nginx-service
Gtk+Message: 15:50:20.604: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```

Step 13: Monitor Cluster using Dashboard

Action:

Open Kubernetes Dashboard

Observe pods, deployments, services, and logs



Dec 19 15:55

Firewall Authentication Keep x Welcome to nginx! x Kubernetes Dashboard x +

http://127.0.0.1:44565/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard/proxy/#/workloads?namespace=default

kubernetes default Search

Workloads

Workloads N

- Cron Jobs
- Daemon Sets
- Deployments
- Jobs
- Pods
- Replica Sets
- Replication Controllers
- Stateful Sets
- Service
 - Ingresses N
 - Ingress Classes
 - Services N
- Config and Storage
 - Config Maps N
 - Persistent Volume Claims N
 - Secrets N
 - Storage Classes
- Cluster
 - Cluster Role Bindings
 - Cluster Roles
 - Events N
 - Namespaces
 - Network Policies N

Pods

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created ↑
hello-cron-29435665-swmhd	busybox	batch.kubernetes.io/controller-uid: 579e4001-9ba6-46c6-a70f-d98320af0db0 batch.kubernetes.io/job-name: hello-cron-29435665 controller-uid: 579e4001-9ba6-46c6-a70f-d98320af0db0 Show all	minikube	ContainerCreating	0	-	-	3 seconds ago
hello-cron-29435664-6c47h	busybox	batch.kubernetes.io/controller-uid: 3440e739-bb0e-4807-9295-ae94725e3d09 batch.kubernetes.io/job-name: hello-cron-29435664 controller-uid: 3440e739-bb0e-4807-9295-ae94725e3d09 Show all	minikube	Completed	0	-	-	a minute ago
hello-cron-29435663-6h69k	busybox	batch.kubernetes.io/controller-uid: 132bb7b9-8be1-4599-a1d2-97a46c286bef batch.kubernetes.io/job-name: hello-cron-29435663 controller-uid: 132bb7b9-8be1-4599-a1d2-97a46c286bef Show all	minikube	Completed	0	-	-	2 minutes ago
hello-cron-29435662-m4f6b	busybox	batch.kubernetes.io/controller-uid: 9afa1841-9e16-4962-a17c-964a611c3195 batch.kubernetes.io/job-name: hello-cron-29435662 controller-uid: 9afa1841-9e16-4962-a17c-964a611c3195 Show all	minikube	Completed	0	-	-	3 minutes ago

Step 14: Cleanup Resources

kubectl delete pods --all